Econometrics 674 **Spring 2025**

Term Project Report
Group 17

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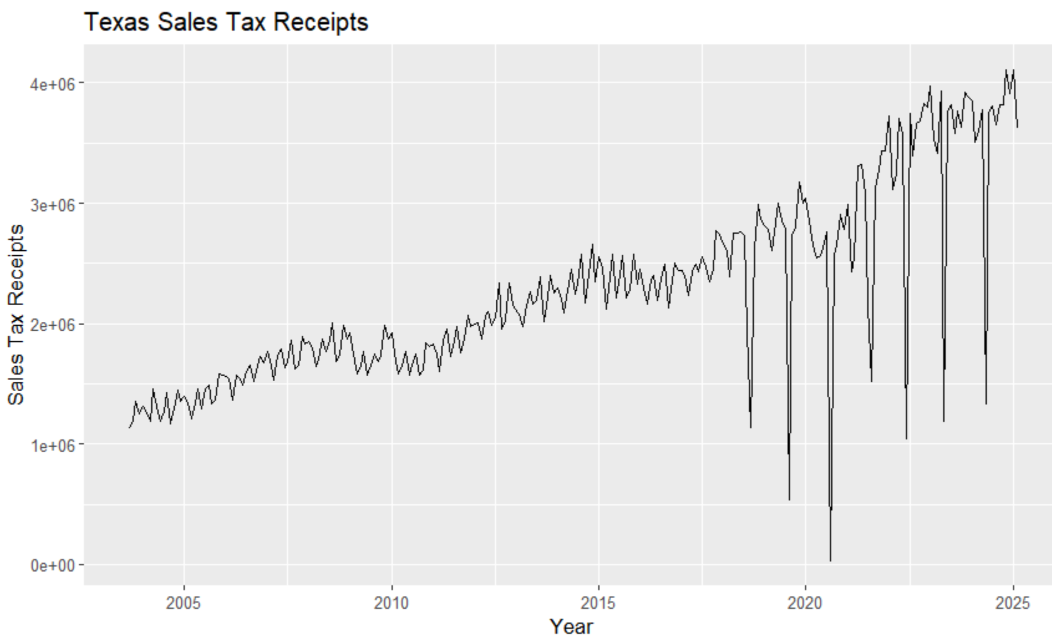
I. Overview of the Sales Tax Variable



I.I Trend Elements

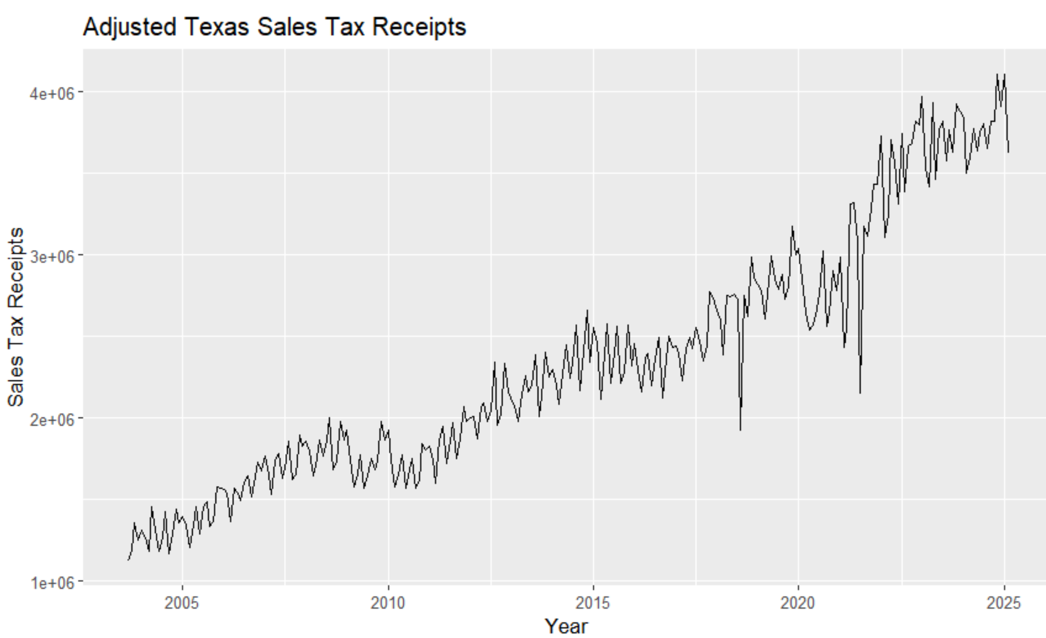
The Unadjusted series graph at the original levels of Texas Sales Tax Receipts from September 2003 to February 2025 displays an overall upward positive trend, indicating growth within Texas's economy. When analyzing the period from around 2004 to 2008, the increase in receipts is very consistent. In 2008, there was a slight break in trend, which made growth ever so slightly plateau for a couple of years following this event. This was mainly caused by the 2008-2009 financial crisis that led to a decline in consumer spending, directly affecting sales taxes collected. After 2010, the economy mostly recovered, and the series resumed its trajectory. However, starting in 2018 and occurring annually, several large drops in sales tax receipts happened yearly due to tax holidays that reduced sales tax collection. In response, we will use an adjusted series, using the log transformation to remove these anomalies from the series, which makes the adjusted series graph follow a more linear trend.

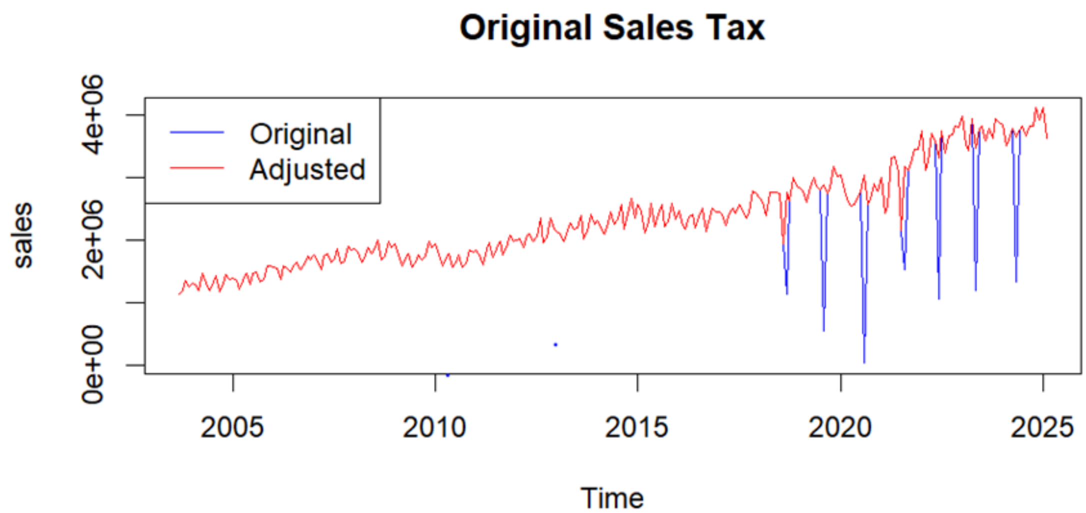
I.II Seasonal Elements

Each fiscal year there is a systematic build up in Texas sales-tax receipts through the first three quarters. This is consistent with potential greater household consumption around spring break, going back to school, or late summer retail. This is followed by an abrupt recurring drop. This shows a sudden collapse in spending . Here we can see that around Covid-19 years there was a drastic decrease in 2020. Thus showing that individuals were not spending as much during this time as many were cautionary against the virus. Beyond the pandemic's distortion, the chart reveals a stable, repeating seasonal cycle whose amplitude scales with state economic growth.

I.III Cyclical Elements

This series also traces broader cyclical fluctuations that mirror statewide and national business cycles. Real growth in sales-tax receipts is shown as the trend line basically flattens after the 2008-2009 Great Recession, and softens again during 2015 during the decrease of oil prices and Texas energy sector busted. The Covid-19 shock also formed the steepest decline as receipts go down in early 2020, overshooting the routine seasonal low. However, the did quickly increase in 2021-2022 as reopening of business, stimulus checks, and an increase in demand fueled consumer spending. By 2023, the series realigned with the pre pandemic trajectory, suggesting that it was back on track. Each of t hese episodes shows up as a plateau or slight decrease in the rising trend line. Overall, this chart documents at least three identifiable downturns and recoveries, each spanning a couple of years and corresponding closely to well-known macroeconomic contractions and rebounds.

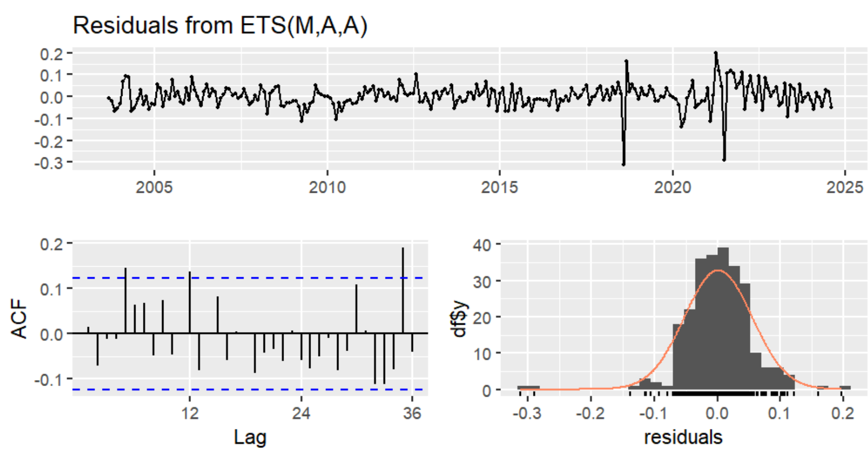


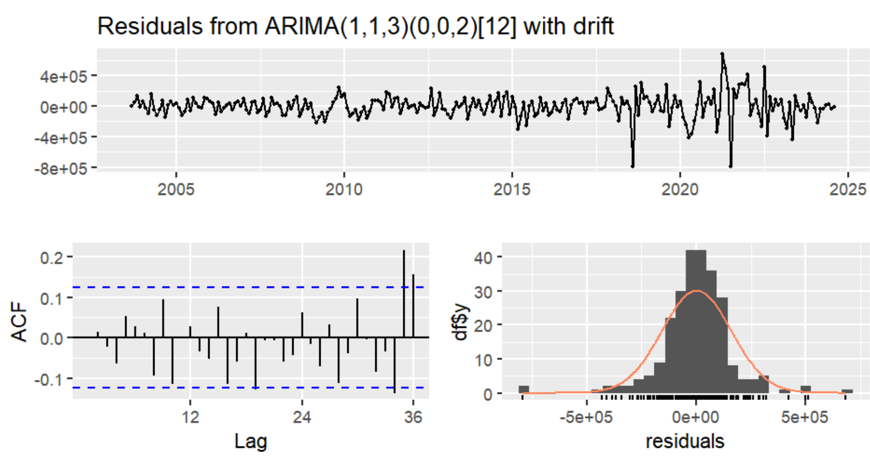


II. Univariate Forecasting Models

II.I ETS and ARIMA

a.i)residual diagnostics





ETS(M,A,A) Model:
The residuals exhibit no clear trend or seasonality and fluctuate around zero, suggesting the model captures the main patterns in the data well. The autocorrelation function (ACF) of the residuals shows that most lags fall within the confidence bands, indicating that the residuals behave like white noise. There is a slight autocorrelation at lag 12, suggesting minor seasonal effects might remain. The residual histogram approximates a normal distribution despite slight left skewness.

ARIMA(1,1,3)(0,0,2)[12] with drift Model:
The residuals fluctuate around zero but show larger variance, especially around extreme periods (e.g., during 2020), indicating potential issues with heteroskedasticity (changing variance). The ACF of the residuals shows some autocorrelation at seasonal lags (12, 24, 36), indicating that some seasonal patterns remain uncaptured. The ACF of the residuals shows some autocorrelation at seasonal lags (12, 24, 36), indicating that some seasonal patterns remain uncaptured.

```
+   Actual = test_data
+ )
  Forecast  Actual
1  3612585 3822535
2  3699237 3822843
3  3942903 4111674
4  3839256 3909853
5  3897122 4107568
6  3732547 3622696
```

ARIMA(1,1,3)(0,0,2)[12] with drift Model:

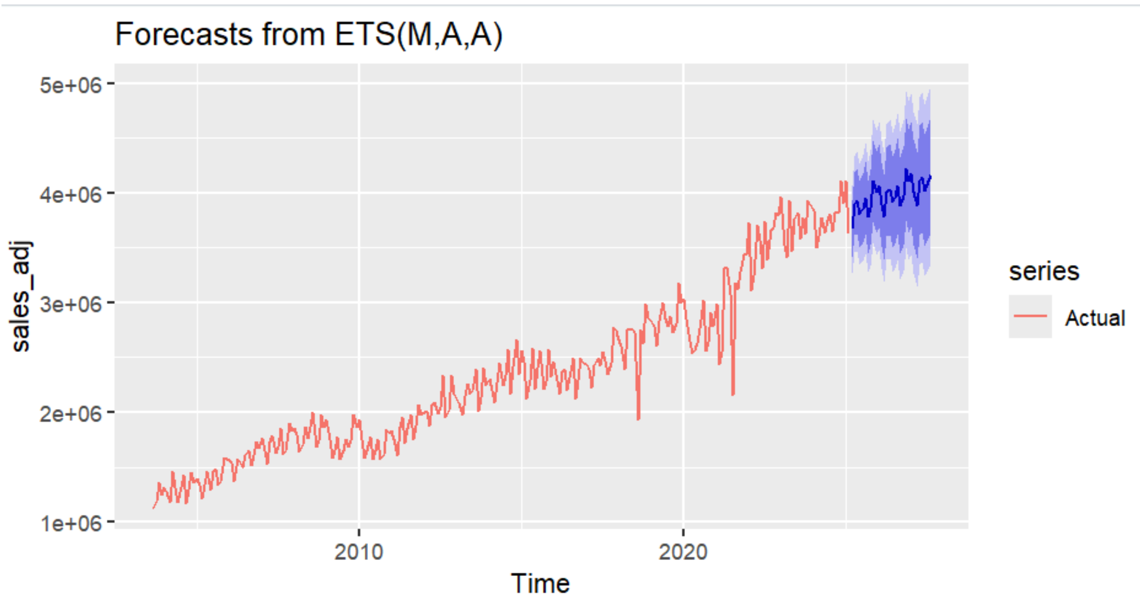
```
  Forecast  Actual
1  3740243 3822535
2  3683465 3822843
3  3792603 4111674
4  3776640 3909853
5  3759188 4107568
6  3660248 3622696
```

iii)

```
> # ETS模型的误差
> accuracy(ets_forecast, test_data)
                     ME     RMSE       MAE        MPE     MAPE
Training set   3460.989 142728.8  91204.68 -0.2029106 3.877859
Test set     112253.130 157681.0 148870.11  2.7878576 3.798624
                  MASE        ACF1 Theil's U
Training set 0.5115574 -0.05681674        NA
Test set     0.8349967 -0.36021527   0.52958
> # ARIMA模型的误差
> accuracy(arima_forecast, test_data)
                      ME     RMSE      MAE        MPE     MAPE
Training set    807.3457 158812.6 108646.2 -0.3378371 4.789765
Test set     164130.3563 211553.5 176647.8  4.0684693 4.413997
                  MASE         ACF1 Theil's U
Training set 0.6093851  0.002684666        NA
Test set     0.9907987 -0.462656765 0.8401012
```

Based on the out-of-sample (test set) performance, the ETS(M,A,A) model outperforms the ARIMA(1,1,3)(0,0,2)[12] with drift model across all key metrics, including RMSE, MAE, MAPE, ACF1, and Theil's U. Therefore, the ETS model is selected as the preferred forecasting model for the Sales Tax variable.

II.II 30 Month Forecasts *Forecasts of Each Month*



```
> ets_forecast_30$mean
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug
2025                 3672556 3900034 3928308 3810328 3862339 3955303
2026 4065502 3898693 3780213 4007692 4035965 3917986 3969996 4062960
2027 4173159 4006350 3887870 4115349 4143623 4025643 4077653 4170617
         Sep     Oct     Nov     Dec
2025 3780793 3867520 4111220 4007591
2026 3888450 3975177 4218877 4115248
2027
```

II.III Comparison of Forecasts

According to Table A-12 of the 2026 - 2027 Biennial REVENUE ESTIMATE, the estimated total sales tax collections for the state 2025 fiscal year were approximately $43,977,083 (thousands), for the 2026 fiscal year $46,211,370 (thousands), and for the 2027 fiscal year was $48,026,569 (thousands). Based on this information, we can determine the accuracy of our 30-month forecast by adding up each monthly projection of the fiscal year to achieve an estimated fiscal year total in order to compare to the BRE estimated forecasted totals. Totals for the 30-month estimated forecast are the following: for the 2025 fiscal year, $46,526,027 (thousands); in the 2026 fiscal year, $47,506,131 (thousands); and in 2027, $48,798,016 (thousands). Based on this information, we can determine that our 30-month forecast is somewhat accurate but tends to overestimate sales tax collections compared to BRE fiscal year projections.

III.Multivariate Forecasting Model

III.I VAR Model of the Sales Tax Variable + 2 more
Two choice variables: Texas Employment and Consumer Price Index (CPI).

Texas Employment (TXNA) – total nonfarm employment in Texas, representing overall economic activity. (Rationale: Higher employment levels typically lead to increased consumer spending, directly affecting sales tax revenues.)

Consumer Price Index (CPIAUCSL) – U.S. Consumer Price Index for all urban consumers, representing inflation. (Rationale: Rising prices (inflation) can increase the nominal value of taxable sales, which also impacts sales tax collections.)

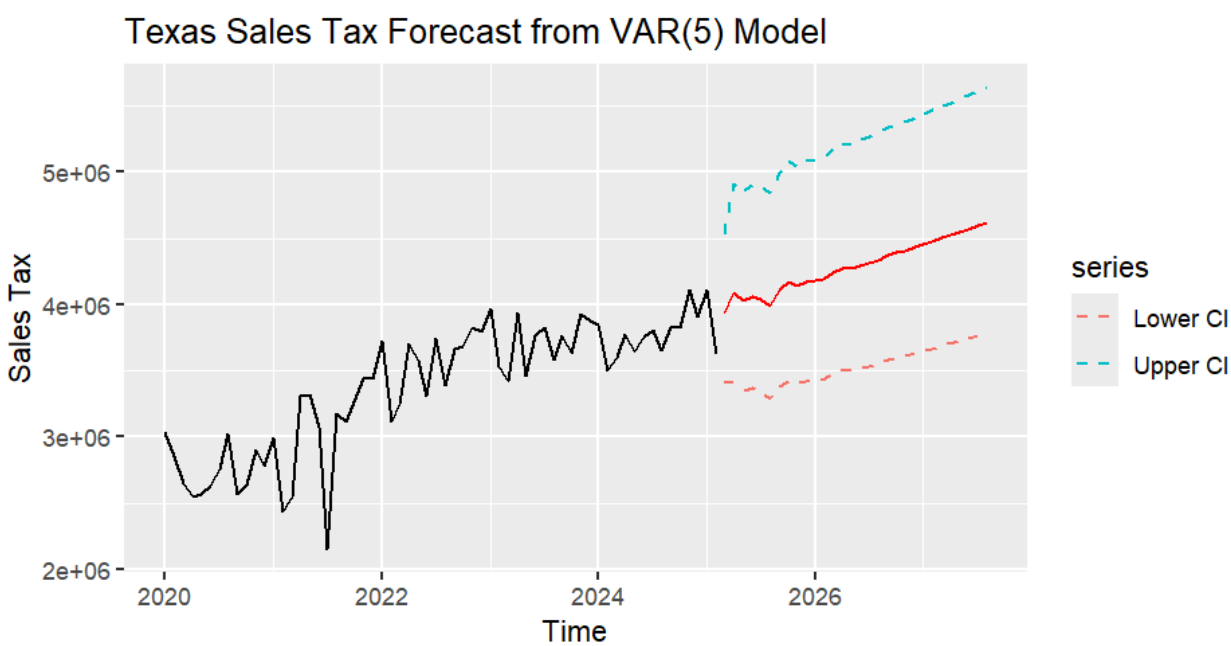
III.II Variable Selection Decision
Texas Employment was chosen to reflect the state's economic strength, as employment directly impacts consumer purchasing power and sales tax receipts. CPI was included to account for inflationary effects. An increase in the CPI can raise the nominal value of sales, thus influencing sales tax collections.

Pre-Testing: Stationarity Tests (ADF Tests): The log-transformed series for Texas Sales Tax, Texas Employment, and CPI were tested using Augmented Dickey-Fuller (ADF) tests; All three series were found to be non-stationary in levels but became stationary after first differencing.

Lag Order Selection: The optimal lag length was determined using the VARselect() function, which considers AIC, BIC, and FPE criteria. Although BIC suggested a lag of 2, VAR(5) was selected based on residual diagnostics (Portmanteau tests), which indicated improved model adequacy with five lags.

III.III 30-Month Forecast

A 30-month ahead forecast for Texas Sales Tax was generated from March 2025 to August 2027 using the VAR(5) model. The forecast demonstrates a steady upward trend in sales tax receipts, reflecting the expected influence of employment growth and inflation.



The ETS(M,A,A) model provided a conservative and stable forecast based solely on the historical behavior of the sales tax variable. The VAR(5) model incorporated macroeconomic variables, allowing it to reflect potential co-movements with employment and inflation, resulting in higher long-term projections but also greater uncertainty.

The results of the VAR(5) model total sales tax collections for the state are the following: $47,553,688 (thousands) for the 2025 fiscal year, $50,687,852 (thousands) in the 2026 fiscal year, and $53,891,427 (thousands) for the 2027 fiscal year. When comparing these results to the Comptroller's forecast, we can determine that this forecasting model tends to overestimate sales tax collections, but the margin decreases yearly. The VAR(5) depicts more linear and consistent growth within the forecast than the ETS(M,A,A), which fluctuates but has an upward trend.